

Practical Circuits

Electric circuit.

1. When will 8 mA of current pass through an electrical circuit

Step 1: $Q = I t$ A) $I = 4/500 = 8 \times 10^{-3} = 8 \text{ mA}$
 $I = \frac{Q}{t}$ c) $I = 8/100 = 80 \times 10^{-3} = 80 \text{ mA}$

2. $Q = I t \rightarrow 5 \times 10^{-3} \times 200 = 1 \text{ C}$

A $\rightarrow$ wrong, that requires energy ($W = qV$)

D $\rightarrow Q = 10 \times 10 = 100 \text{ C}$, not 1 C

~~Question 2~~ Answer C.

3. $t = 3 \text{ min} = 1800 \text{ s}$, $I = 10 \text{ mA} = 0,01 \text{ A}$

$Q = I t = 0,01 \times 1800 = 18 \text{ C}$.

I have RV

The Ampermeter is in series, so it always measures the current (correct) regardless of its resistance - B and D are irrelevant ✓

* Q4 1.

Resistance is defined by Ohm law:

$V = IR \rightarrow$ potential difference = current $\times$ resistance ✓
The other options use resistance in ~~the~~ different formulas - energy, power, sensitivity - but none of them define resistance itself.

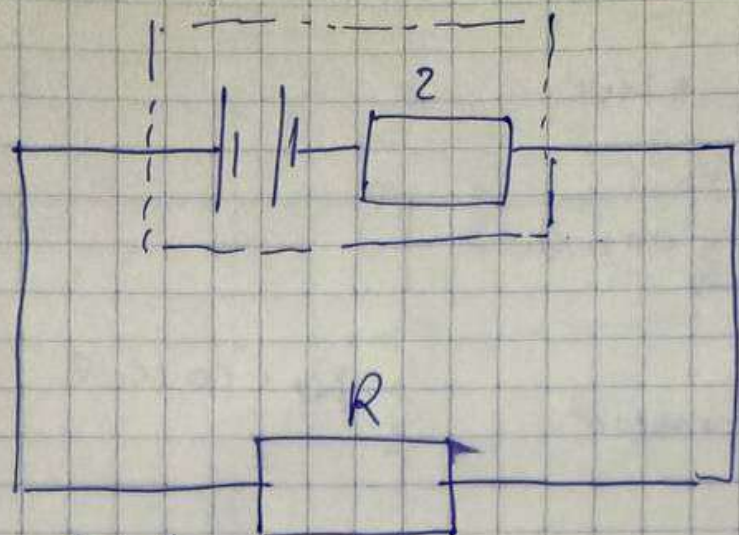
* 2 $R = \frac{V}{I}$ $V = \frac{E}{Q}$

$$V = \frac{E}{Q} = \frac{\frac{1}{2}mv^2}{It} = \frac{[\text{kg m}^2 \text{s}^{-2}]}{[\text{A s}]} = \text{kg m}^2 \text{s}^{-3} \text{A}^{-1}$$

So, the SI base units of resistance are

$$\bullet R = \frac{V}{I} = \frac{[\text{kg m}^2 \text{s}^{-3} \text{A}^{-1}]}{[\text{A}]} = \text{kg m}^2 \text{s}^{-3} \text{A}^{-2}$$

* 4



Step 1: Find current

$$t = 2 \text{ min} = 120 \text{ s}$$

$$I = Q/t = 240/120 = 2 \text{ A}$$

Step 2: Find resistance R

$$R = E/I^2 t = 1440/(2^2 \times 120) = 1440/480 = 3.0 \Omega$$

Step 3: Find total resistance

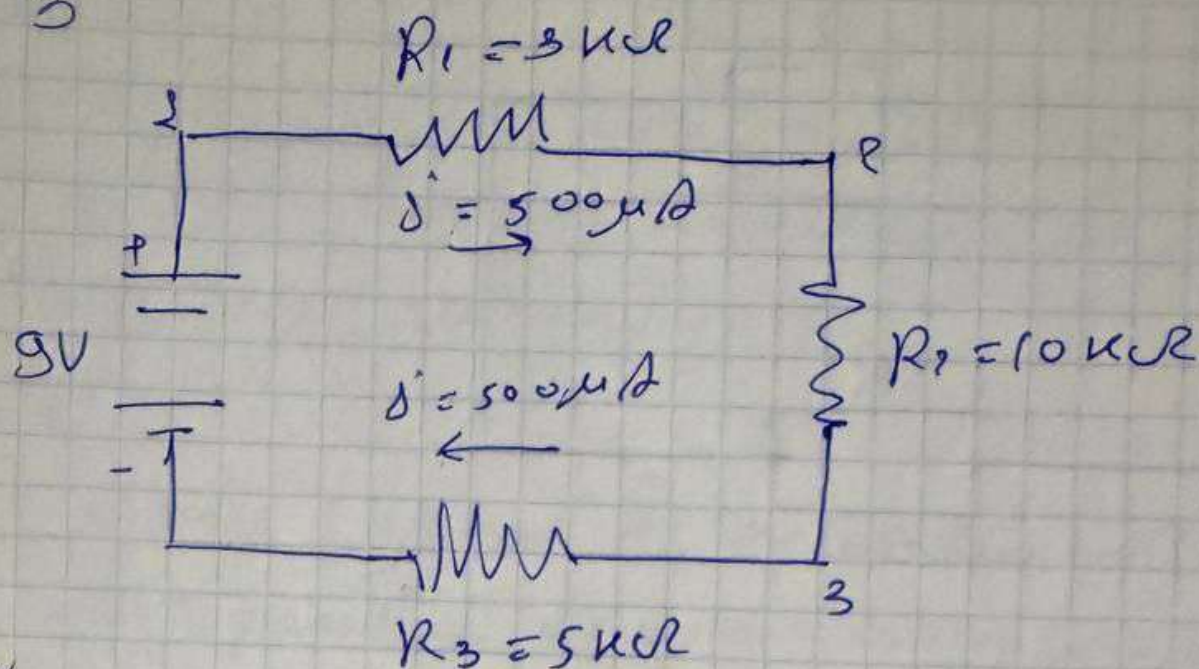
$$R + r = E/I = 9.0/2 = 4.5 \Omega$$

Step 4: Find internal resistance?

$$r = 4.5 - 3.0 = 1.5 \Omega \checkmark$$

* The voltmeter is connected in parallel with R. If R and R_v are similar in size, current will flow through the voltmeter too - this gives wrong readings for an accurate measurement, the voltmeter must take almost no current. This only happens when R_v is much larger than R - meaning R is smaller

* 3



Known:

- $I = 500 \mu A = 500 \times 10^{-6} A$
- $R_1 = 3 k\Omega$, $R_2 = 10 k\Omega$, $R_3 = 5 k\Omega$

Voltage drop across each resistor:

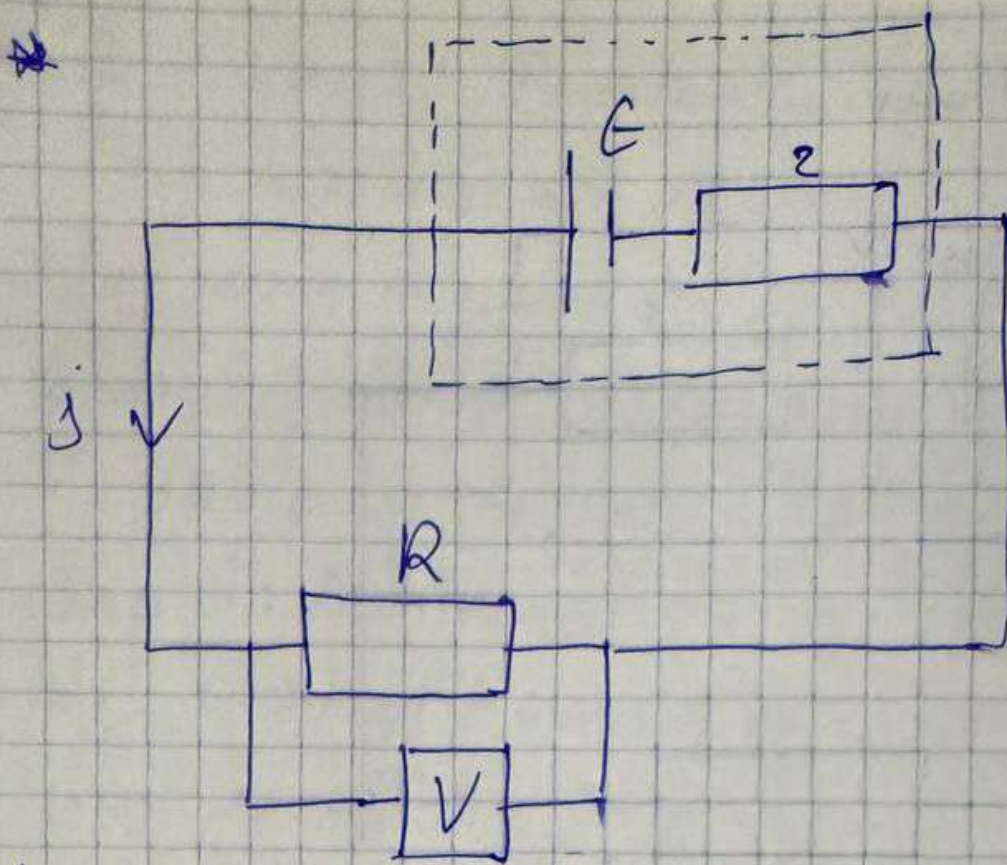
$$V_1 = I \times R_1 = 500 \times 10^{-6} \times 3 \times 10^3 = 1.5 V$$

$$V_2 = I \times R_2 = 500 \times 10^{-6} \times 10 \times 10^3 = 5.0 V$$

$$V_3 = I \times R_3 = 500 \times 10^{-6} \times 5 \times 10^3 = 2.5 V$$

Check - Kirchhoff's Second Law

$$V_1 + V_2 + V_3 = 1.5 + 5.0 + 2.5 = 9.0 V = \text{battery voltage}$$



Known:

- $I = 0,6 \text{ mA} = 0,6 \times 10^{-3} \text{ A}$
- $R = 2,5 \text{ k}\Omega = 2,5 \times 10^3 \Omega$

Step 1: Convert units.

$$I = 0,6 \times 10^{-3} \text{ A}$$

$$R = 2,5 \times 10^3 \Omega$$

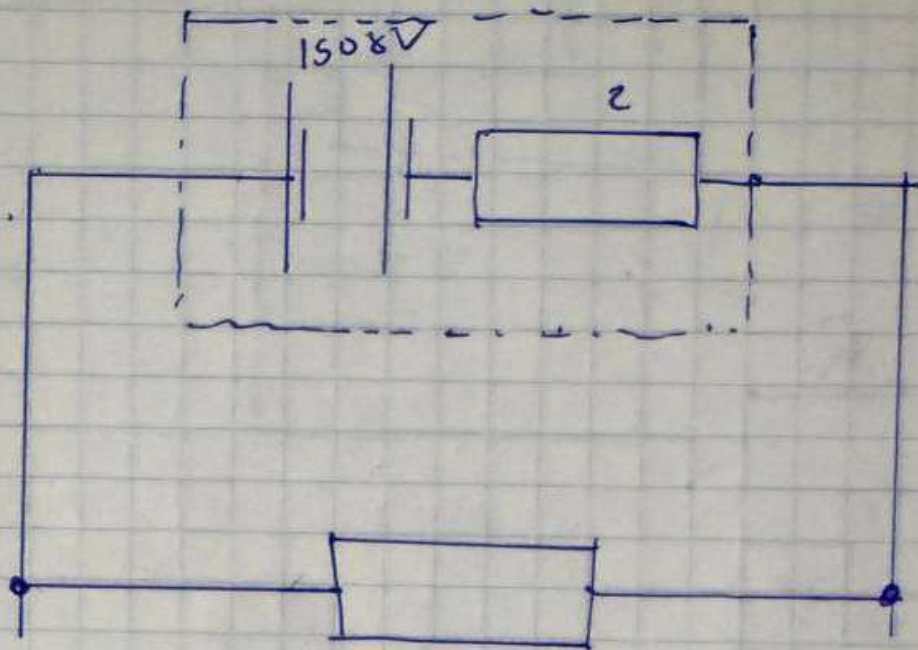
Step 2: Calculate V

$$V = IR = (0,6 \times 10^{-3}) \times (2,5 \times 10^3)$$

$$V = 1,5 \text{ V}$$

to the same battery

*



Known:

- $\mathcal{E} = 1508 \text{ V}, I = 0,4 \text{ A}, P = 600 \text{ W}$

Step 1: Find V across resistor A

$$P = IV \rightarrow V = P/I = 600/0,4 = 1500 \text{ V}$$

Step 2: Find internal resistance r

$$\mathcal{E} = V + Ir \rightarrow r = (\mathcal{E} - V)/I$$

$$r = (1508 - 1500)/0,4 = 8/0,4$$

$$r = 20 \Omega \checkmark$$

* 1. Step 1: Resistance of the wires

$$\text{Each wire} = 800 \text{ m} \times 0.0050 = 4 \Omega$$

$$\text{Two wires in series} \rightarrow R_{\text{wires}} = 4 + 4 = 8 \Omega$$

Step 2: Voltage drop across the wires

$$V = IR = 0.00 \times 8 = 4.8 \text{ V}$$

Step 3: Minimum e.m.f

By Kirchhoff's ~~sec.~~ second law — Total e.m.f =
= sum of all p.d in the circuit:

$$\text{em.f} = V_{\text{battery}} + V_{\text{wires}} = 16.0 + 4.8 = 20.8 \text{ V}$$

* 2. Voltage $\rightarrow$ Volts (V)

Current — Amperes (A)

Resistance — ohms (Ω)

* 3. The three bulbs are connected in parallel to a 6V battery.

In parallel circuit, each branch receives the full 6V — the same as the battery voltage.

So each 6 volt bulb receives exactly 6V — the same as if a single bulb were connected alone. Therefore all three bulbs shine at the same time brightness as a single bulb connected

Kirchhoff's Second Law II

Statement: the total e.m.f in a closed circuit is equal to the sum of the potential differences across all components.

Formula: $\mathcal{E}_1 + \mathcal{E}_2 = V_1 + V_2$

Key points:

- This is a consequence of conservation of energy
- Energy transferred into the circuit = energy transferred out
- In a series circuit, the total p.d is split across components depending on their resistance.
- In a parallel circuit, the p.d is the same across each closed loop.
- Each closed loop as its own independent series circuit
- If one branch breaks, current still flows through the other branches.

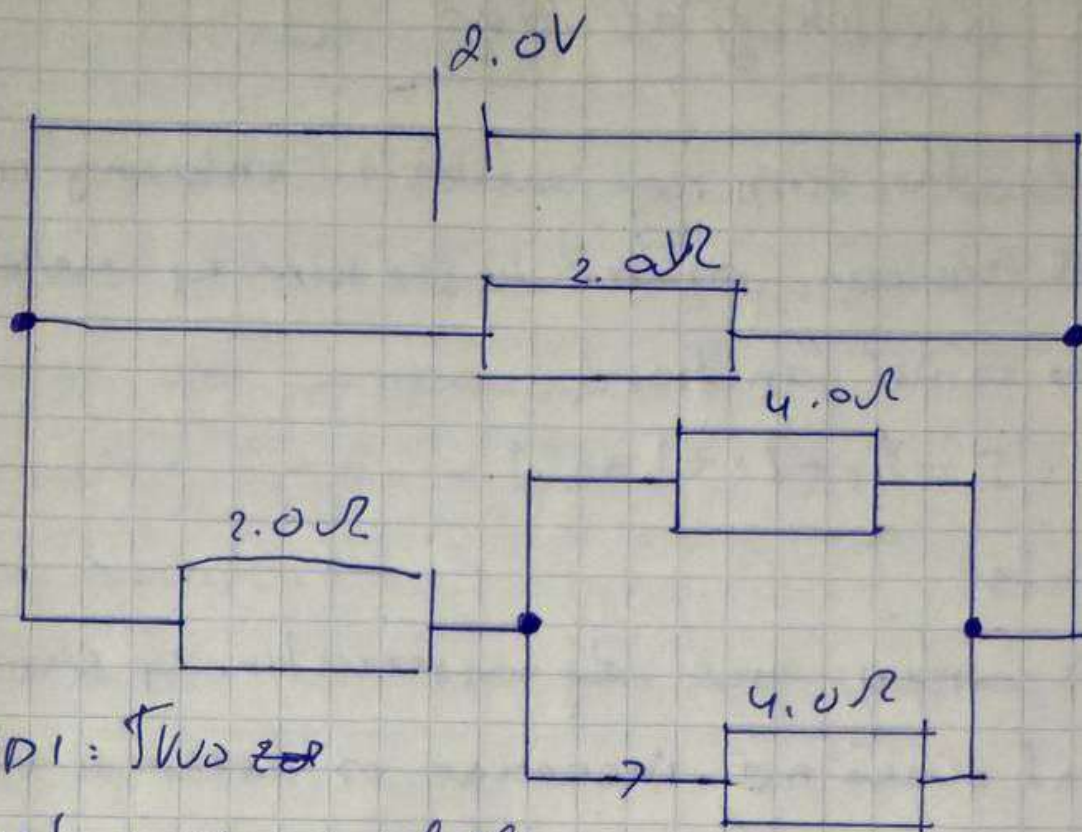
Kirchhoff's laws I

Statement: the sum of currents entering a junction is always equal to the sum of currents leaving the same junction.

Formula: $I = I_1 + I_2 + I_3$

Key points:

- This is consequence of conservation of charge
- Current does not increase or decrease when it splits at a junction
- A junction is a point where at least three circuit paths meet.
- A branch is a path connecting two junctions.
- In a series circuit, current is the same at every point.
- In a parallel circuit, current divides at junctions.
 - each branch carries different value.



Step 1: Two ~~2.0~~

resistors in parallel.

$$1/R = 1/4 + 1/4 = 2/4 \rightarrow R = 2\Omega$$

Step 2: Add 2Ω in series

$$R = 2 + 2 = 4\Omega$$

Step 3: This 4Ω parallel with 2Ω (top branch)

$$1/R_{\text{total}} = 1/4 + 1/2 + 2/4 = 3/4 \rightarrow R_{\text{total}} = 4/3\Omega$$

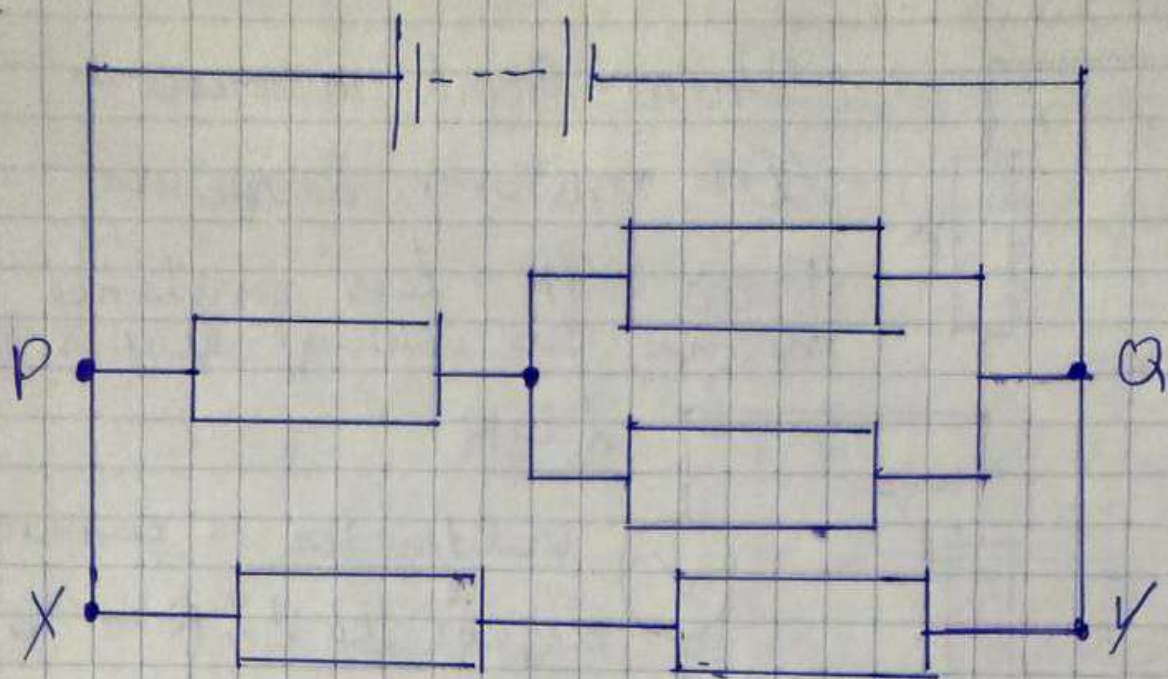
Step 4: Total current from cell.

$$I = V/R = 2.0 / (4/3) = 1.5\text{ A}$$

Step 5: Current I through the marked branch
Three equal 2Ω branches $\rightarrow$ current splits
equally:

$$\text{Each branch} = 2/6 \times 1.5 = 0.5\text{ A}$$

The splits equally between two 2Ω : $I = 0.5/2 = 0.25\text{ A}$



Between P and Q there are 2 resistors in parallel, not in series. In parallel, total resistance is less than each individual resistor, so the total resistance is not equal to the sum - C is false V.

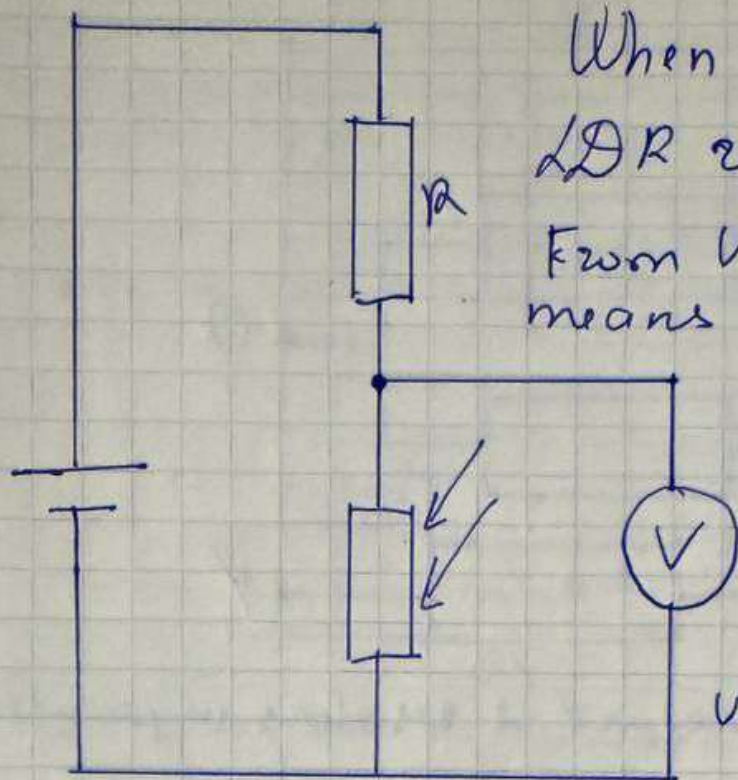
Why others are true:

A - PQ and XY are parallel branches $\rightarrow$ same V.

B - X to Y has two resistors in series $\rightarrow R_T = R_1 + R_2$

D - PQ has parallel resistors $\rightarrow$ less total resistance than XY V.

★



When light increases $\rightarrow$
 LDR resistance decreases.
 From $V=IR \rightarrow$ less resistance
 means less voltage across the
 LDR.

Voltmeter is connected
 across the LDR $\rightarrow$ so
 voltmeter reading decreases
 ✓.

Voltage divide Circuit.

The combined resistance R_T of two resistors of resistances R_1 and R_2 connected in parallel is given by the formula.

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

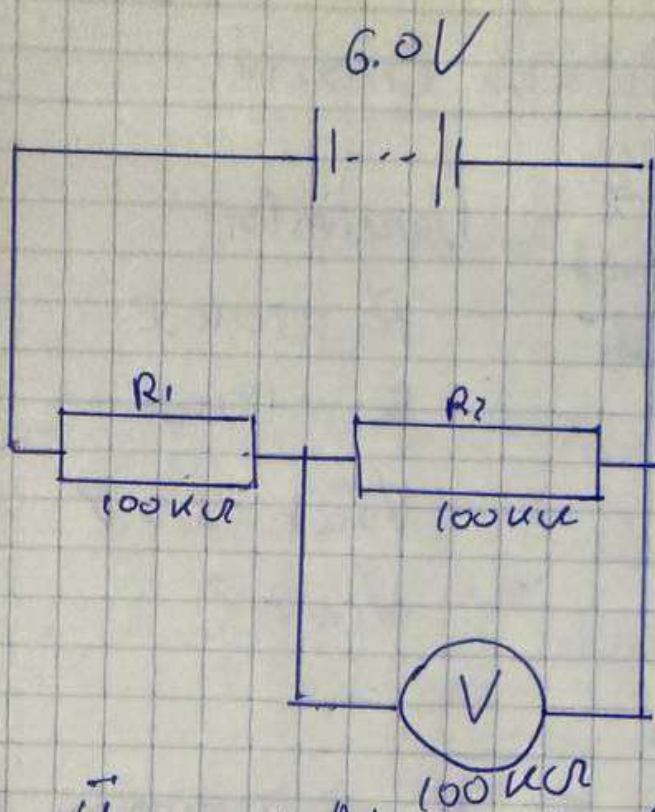
Which statement is used in the derivation of this formula?

In parallel circuit, both resistors are connected ~~across~~ across the same supply. Therefore the potential difference is the same - this is the ^{key starting point}

$$I_1 = V/R_1 \text{ and } I_2 = V/R_2$$

$$\text{Total current; } I = I_1 + I_2 = V/R_1 + V/R_2 = V(1/R_1 + 1/R_2)$$

from Ohm's law; $V/I = R_T$, therefore: $1/R_T = 1/R_1 + 1/R_2$
 ✓



Step 1. R_1 vs Voltmeter
parallel müqaviməti?
 $R_2 = 100\text{k}\Omega$, Voltmeter =
 $100\text{k}\Omega$.
parallel qarale
 $1/R_{\text{parallel}} = 1/100 + 1/100 =$
 $= 2/100$
 $R_{\text{parallel}} = 50\text{k}\Omega$.

Step 2. Ümumi Müqavimət.

R_1 seriyada, parallel hissə seriyada:

$$R_{\text{total}} = 100 + 50 = 150\text{k}\Omega$$

Step 3: Ümumi cərəyan

$$I = V/R = 6.0 / 150000 = 40\mu\text{A}$$

Step 4. R_2 -dəki cərəyan,

$40\mu\text{A}$ R_1 dan keçir. Sonra R_2 vs Voltmeter arasındakı bölünür.

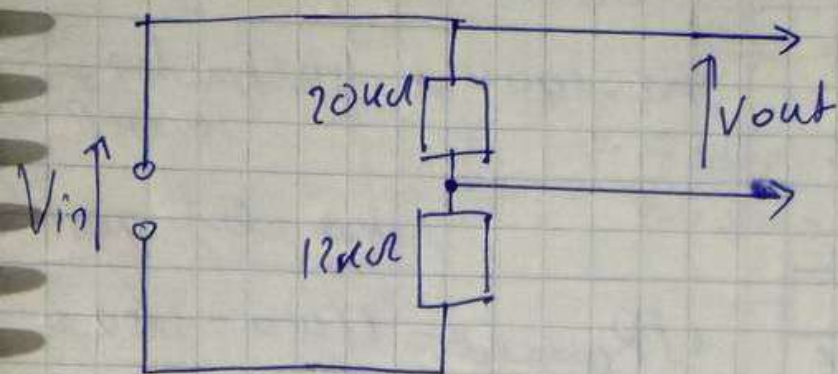
— Hər ikisi eyni müqavimətdədir? — $100\text{k}\Omega$.

Ona görə cərəyan bərabər bölünür.

$$I(R_2) = 40/2 = 20\mu\text{A}. \checkmark \quad \text{Nəticə: } 20\mu\text{A}.$$

★

Potential divider Circuit.



Verilenler:

$$R_1 = 20k\Omega$$

$$R_2 = 12k\Omega$$

$$V_{out} = 5,3V$$

$$V_{in} = ?$$

Step 1 : Formula

$$V_{out} = R_2 / (R_1 + R_2) \times V_{in}$$

Verilen R_2 nin üzerindeki değer, R_1 -in üzerindeki değere eşittir.

Ona göre, R_1 numaratörüne yazılır.

Step 2 : V_{in} için genellenen düzleşt.

Her iki tarafı $R_1 / (R_1 + R_2)$ -ye böldükten:

$$V_{in} = (R_1 + R_2) / R_2 \times V_{out}$$

Step 3: Değeri yerleştir.

$$V_{in} = (20 + 12) / 20 \times 5,3$$

$$V_{in} = 32 / 20 \times 5,3$$

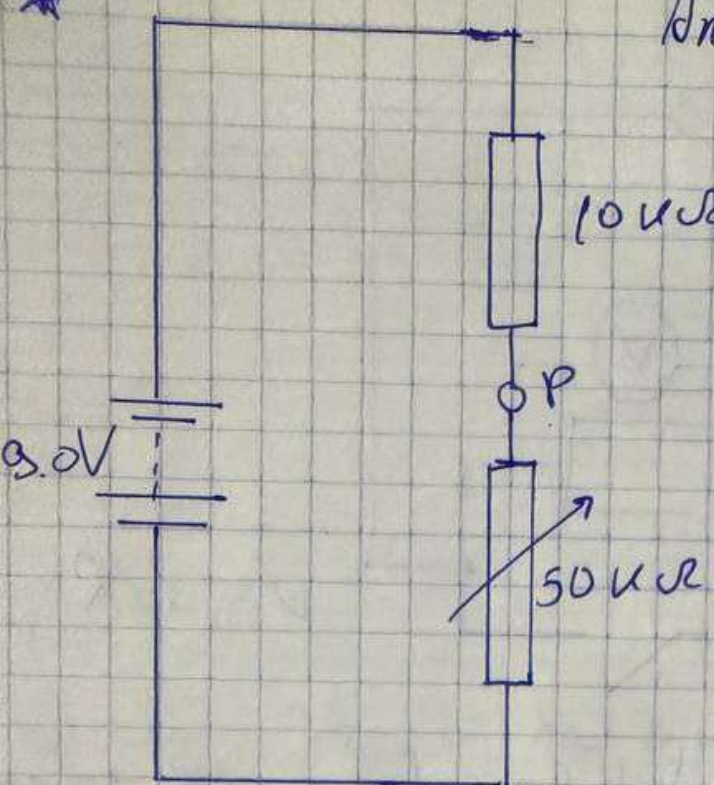
$$V_{in} = 1,6 \times 5,3$$

$$V_{in} = 8,5V$$

Yoxlama:

$$V_{out} = 20 / (20 + 12) \times 8,5 = 20 / 32 \times 8,5 = 0,625 \times 8,5 = 5,3V$$

*



Answer: B zero to 7.5 V.

Formula: $V_2 = V_{in} \times R_1 / (R_1 + R_2)$

$10k\Omega$ $R_1 = \text{variable } (0 \text{ to } 50k\Omega)$

$R_2 = 10k\Omega, V_{in} = 9V$

Minimum $\rightarrow R_1 = 0$

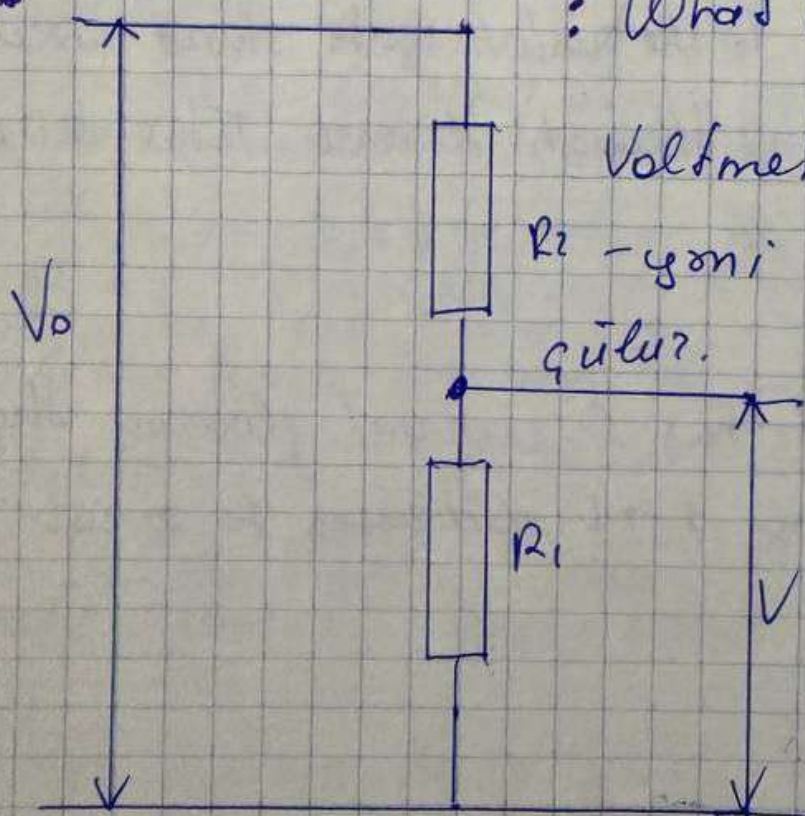
$V_2 = 9 \times 0 / (0 + 10) = 0V$

Maximum $\rightarrow R_1 = 50k\Omega$

$V_2 = 9 \times 50 / (50 + 10) = 9 \times 50 / 60 = 7.5V$

Range = 0 V to 7.5V.

*



: What is the value of V?

Voltmeter R_1 - in üzründədir.

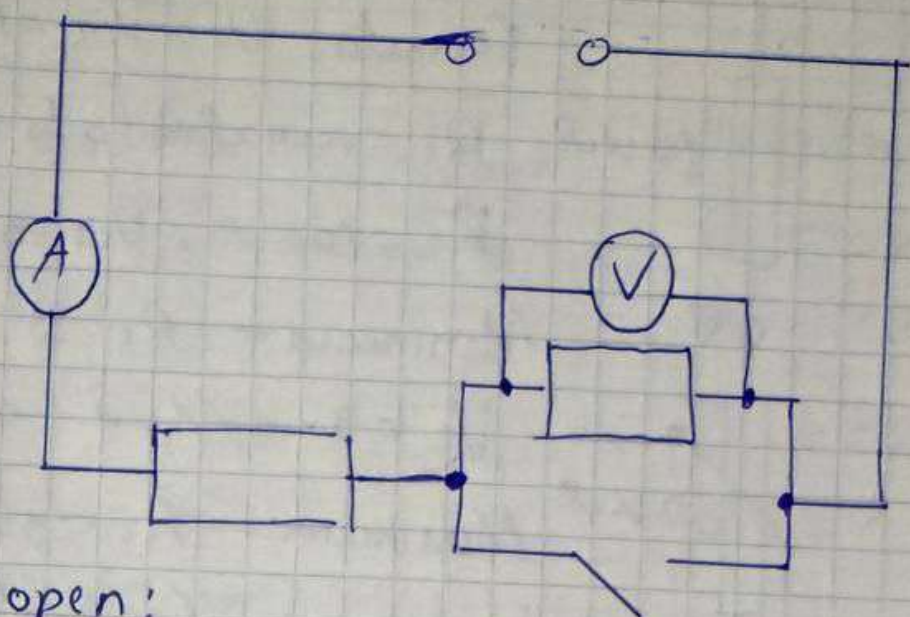
R_2 - gəni R_1 -dən gərginliyi ölçür.

Formula:

$V = V_0 \times R_1 / (R_1 + R_2)$

C-olur R_2 var - amma voltmeter R_1 -nin üzründə deyil.

R_1 -in üzründədir. Ona görə (B).



$$V = I/R$$

C

Switch open:

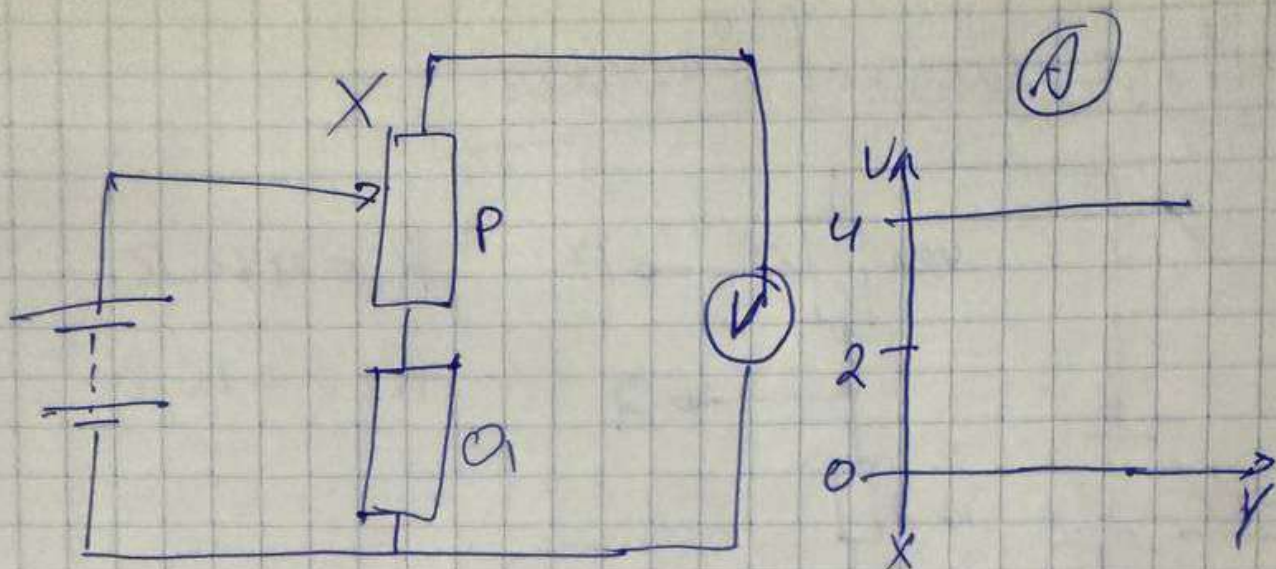
- two resistors in series $\rightarrow$ total resistance $= 2R \rightarrow$ current $\downarrow$
 $E/2R$

Switch closed:

- the resistor parallel to the switch gets short-circuited - all current bypasses it through the wire. Total resistance $= R \rightarrow$ current $= E/R$

So I increases. ✓

the short circuit resistor has 0 current flowing through it $\rightarrow$ 0 voltage across it $\rightarrow V$ decreases to zero. ✓



Ans: A.

Because: the V has a very high resistance. This means almost no current flows through Q at any point.

— When the slider is at X — the potentiometer resistance is zero. Current flows directly through the V . Voltmeter reads $4V$.

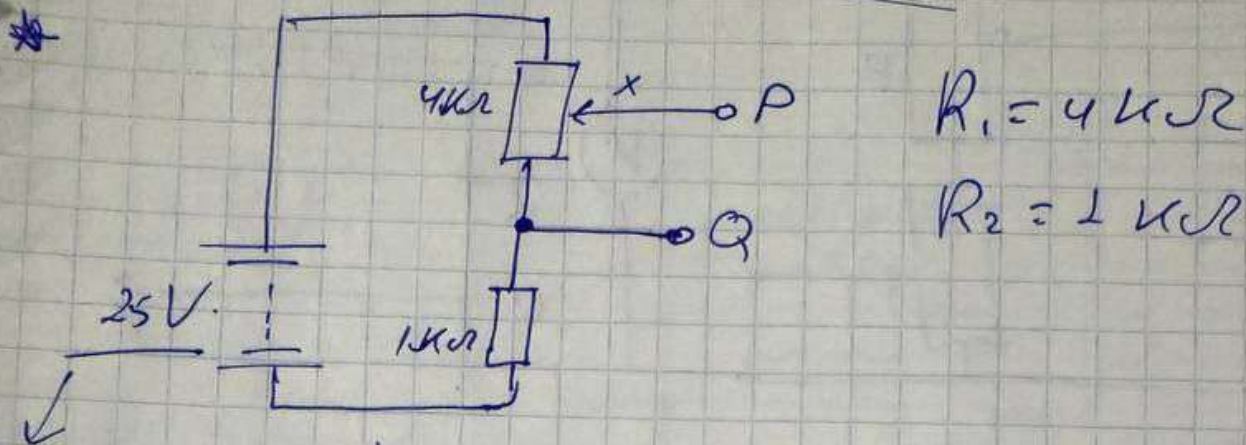
— When the slider moves to Y — potentiometer P and the Voltmeter are in series. But because the voltmeter resistance is so high, almost all the voltage drops across the voltmeter.

Voltmeter still reads $4V$

So as the slider moves from X to Y — the Voltmeter reading stays constant at $4V$

That is why the graph is straight horizontal line at $4V$.

Voltage Current Divider



Battery - here, asked. P that's why we will write P at the top of the formula.

So, $V_{out} = V_{in} \times R_1 / (R_1 + R_2)$

$V_2 =$ is a Voltage between P and Q

$V_{avg} = \text{battery } V \rightarrow 25V$

R_1 - changeable resistor between $0-4$.

$R_2 = 1\text{ k}\Omega$.

$V_{avg} = 25V$

$V_{max} \rightarrow R_1 \rightarrow 4 \quad V_2 = 25 \cdot 4 / (1+4) = 20V$

$V_{min} \rightarrow R_1 \rightarrow 0 \quad V_2 = 25 \cdot 0 / (1+0) = 25 \cdot 0 = 0V$

$V_2 = 0V \longleftrightarrow 20V$

$Q = i \cdot t$ L1 Electrons & Charge (Elektronlar vs yük)
 conduction electrons - metallarda, elektronların atom şarjına
 hırsızları
 conventional current - realde - $\rightarrow +$, ama direkler $+ \rightarrow -$

charge (C) $\rightarrow Q = i \cdot t \rightarrow \frac{\text{time (s)}}{\text{load}}$

$\downarrow$
 Current (A)

Q - ~~sonuçta neş electron~~ ^{carşyon} ~~vesdığını gösterir. comi neş electron~~
 t - ne qum miadad ordı

i - sonıqırs neş electron kesir (axın sürüş)

Q - comi' neş electron kesir

L2 Potential Difference (Göçinlik) V

$V = W / Q$ $\frac{\text{yük (C)}}{\text{charge}} \rightarrow$ bir coulomb'a dırsın energy.

$\downarrow$ Volt (V) $\downarrow$ Energy $\downarrow$ ~~muqavimst~~ (electronı itelırsın qumı).
 Göçinlik $\uparrow$ Voltage Göçinlik.

Ohm $\rightarrow V = i R \rightarrow \frac{\text{Muqavimst}}{\text{ne qumı ofomla toplanır.}}$

$\downarrow$
 Carşyon (Ampere)

sonıqırs neş electron kesir.

EMF vs PD

Same: her ikisi Voltla ölçülür. $V = \frac{W}{Q}$

EMF: Bataryanın vermesi' energy - max verim Q

PD = Rezistans/dampının aldığı energy - max verim.

L3 Series and Parallel Circuits

1 Series

$i_1 = i_2 = i_3 \rightarrow$ carşyon aynı

$V_1 + V_2 + V_3 = V_{\text{total}} \rightarrow$ ygöçinlik bölünür

$R_{\text{total}} = R_1 + R_2 + R_3 \rightarrow$ Muqavimst toplanır

2 Parallel

$V_1 = V_2 = V_3 \rightarrow$ Göçinlik her yerdə aynıdır

$i_{\text{total}} = i_1 + i_2 + i_3 \rightarrow$ Carşyon bölünür

$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

4 EMF, Internal Resistance, Circuit Symbols

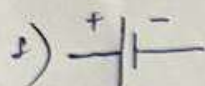
Gerginlik, bir coulomb dâim enerji iletiriz (V, E (EMF))

$$V = IR \rightarrow \text{(PD) zomı yaranan gerginlik (enerji itgisi)}$$

$$E = I(R+r) \rightarrow \text{Bataryanın iç direnci den enerji (Rezistor + Batary)}$$

$$\text{Lost volts} = Ir \rightarrow \text{Bataryanın içinde den enerji}$$

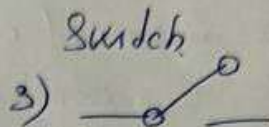
Circuit Symbols



Batary / Cell



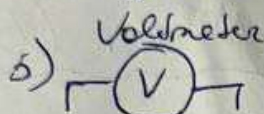
Corayonı Mshduklazdır



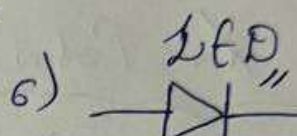
Dövrni açar / bağlayıcı



Series gaxılır



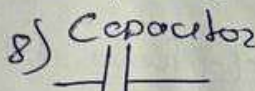
Paralel gaxılır



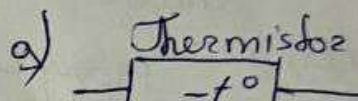
İşık iletir + ısıq



Müqavimət dəyişir



İki saxlayıcı



İstilik → müqavimət azalır



İşıq → müqavimət azalır



fərq iletir + keçirir



İşıq + istilik verir

$$R_1 = 2\Omega \quad R_2 = 4\Omega \quad R_3 = 6\Omega$$

$$V = 12V$$

series

$$\text{total } I = 1A$$

$$R_1 = 3\Omega \quad R_2 = 4\Omega \quad R_3 = ?$$

$$EMF = 12V \quad 2\Omega \text{ R. } V = 10V \quad S = ?$$

$$12 = I(R+r) \Rightarrow 12 = 10 + I \cdot 1 \quad S = 2$$

$$S = 2 \quad R = 4$$

$$2\Omega \text{ R. } R = 2\Omega \quad 3 = 5 \cdot 4$$

$$S = 1$$

$$Q = 60C$$

$$I = 120 \text{ s.m}$$

$$I = \frac{60}{120} = 0.5$$